

The Next-Generation School Safety & High-Speed Attendance Ecosystem
Secure Access Gateway for Student Safety • Powered by Eastern Automations

Business Requirement Document (BRD)

Document Title	KidShield v2.0 — Business Requirement Document
Version	1.1 (Aligned with Deployed Live Site)
Document Owner	Product Management, KidShield Technologies
Document Status	Approved for Stakeholder Review
Date	02 August 2026
Classification	Confidential

Prepared for: School Administrators, Principals, Teachers, Parents/Guardians, and KidShield Internal Technology Teams.

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1. Executive Summary

KidShield v2.0 is a digital school-safety and attendance platform that replaces traditional paper-based roll calls and disconnected parent communication with a high-speed, QR-code-driven ecosystem. Each student is issued a cryptographically unique smart tag (QR code) affixed to their school bag. Scanning this tag with any smartphone instantly performs the intended action — recording attendance, routing a parent-teacher meeting request, identifying the owner of a lost item, or displaying emergency contact information to a good Samaritan.

The platform is delivered as a responsive, installable Next.js web application (Progressive Web App) comprising five role-based portals — Parent Portal, Staff/Teacher Hub, Principal Terminal, School Admin Command Center, and a restricted Overseer (HQ) console. The system is branded as a Secure Access Gateway for Student Safety and is powered by Eastern Automations. Data is persisted in a cloud-hosted Supabase (PostgreSQL) database with authentication and role-based access controls. The deployed production environment (kidshield-v2.vercel.app) adds PWA installability, real-time data synchronization on dashboards, and a system-status panel. This document specifies the business requirements governing the design, build, and acceptance of the system.

2. Project Background & Business Context

Schools today face several operational pain points that directly affect child safety and institutional efficiency:

- **Slow & error-prone attendance:** Manual roll calls consume 5–10 minutes per class and are prone to proxy marking and recording errors.
- **Delayed parent notification:** There is no real-time channel to inform parents when a child does not arrive at school or when an emergency occurs.
- **Inefficient parent-teacher communication:** Meeting invitations are paper-based, easily lost, and offer no delivery confirmation.
- **Lost belongings recovery:** Unlabeled school items are difficult to return to their owners, creating administrative burden.
- **Emergency data not accessible:** In an emergency, bystanders and first responders have no safe way to reach a child's parent or access critical medical information.

KidShield v2.0 addresses these challenges with a single, unified ecosystem built around a durable QR tag and a set of purpose-built web portals.

3. Business Objectives & Success Metrics

3.1 Strategic Objectives

- 1 Reduce daily attendance-taking time per class from ~8 minutes to under 60 seconds.
- 2 Provide parents with near-real-time visibility and alerts regarding their child's school-day safety.
- 3 Digitize and streamline parent-teacher meeting (PTM) invitation delivery with delivery confirmation (app push or WhatsApp fallback).
- 4 Establish a campus-wide lost-and-found identification service using the same smart tag.
- 5 Create an emergency-response layer that lets any bystander safely contact a child's parent when a tag is scanned.
- 6 Equip school leadership (Principal & Admin) with live attendance analytics and alerting dashboards.

3.2 Key Success Metrics

Metric	Target	Measurement Source
Attendance processing time	≤ 60 seconds per class	Admin dashboard timers / logs
Attendance accuracy	≥ 99% (elimination of proxy marking)	Reconciliation audits
Notification delivery rate	≥ 98% (app push + WhatsApp fallback)	Meeting request logs
Parent onboarding completion	≥ 90% of enrolled students	Registration records
Tag activation turnaround	Same-day activation	Tag provisioning logs
Lost item resolution time	≤ 24 hours	Lost & found logs

4. Stakeholders & User Roles

4.1 Stakeholder Register

Role	Primary Business Responsibility
Parents / Guardians	Register children, activate tags, receive alerts & PTM invites, view emergency profiles.
Teachers / Staff	Take attendance, trigger PTM requests, manage lost & found, export attendance reports.
Principal / Campus Admin	Monitor campus-wide attendance, review failed alerts, manage class sections, analytics, license renewal.
School Administrator (Admin)	Manage staff assignments, view live campus analytics, emergency hub, export master CSV.
Overseer / HQ (Super Admin)	Batch-generate tag batches, global fleet analytics, master database access (CRUD).
Bystander / Good Samaritan	Scan a lost child's tag to view safe contact actions (call / WhatsApp the parent).
KidShield Product & Engineering	Build, operate, and maintain the platform and tag supply chain.

4.2 Role-Based Access Matrix

Capability	Parent	Teacher	Principal	Admin	Overseer
Main Portal	R	R	R	R	R
Parent Registration	W	–	–	–	–
Take Attendance (QR scan)	–	W	–	–	–
PTM Request & Lost & Found	–	W	R	R	–
Attendance CSV Export	–	W	R	R	–
Live Analytics Overview	–	–	R	R	R
Staff Assignment	–	–	–	W	R
Emergency Hub	–	–	R	W	R
Tag Batch Generation	–	–	–	–	W

Capability	Parent	Teacher	Principal	Admin	Overseer
Fleet / Global Analytics	–	–	–	–	R
Master DB Vault (CRUD)	–	–	–	–	W

R = Read, W = Write/Execute, – = No Access. The Overseer (HQ) console additionally requires Level-5 clearance authentication before any capability is exposed.

5. Scope of the System

5.1 In-Scope

- QR-based student smart tag lifecycle: batch generation, provisioning, activation, and archive.
- Parent self-registration portal with OTP phone verification and child profile creation.
- Teacher classroom command hub with batch QR scanning for attendance, PTM invites, and lost & found.
- WhatsApp / push-notification fallback routing for parent alerts.
- Principal terminal with class-wise attendance, failed-alert management, analytics, and system setup.
- Admin command center with live campus overview, staff assignment, and emergency hub.
- Overseer (HQ) console with restricted authentication, tag forging, fleet analytics, and DB vault.
- Bystander emergency scan page with verified child details and secure contact actions.
- Attendance CSV export for daily records.
- Public privacy policy page aligned with Indian data-protection law.

5.2 Out-of-Scope (for v2.0)

- Native mobile applications (current delivery is responsive web only).
- Hardware/embedded RFID readers; the v2.0 tag is a printed QR sticker.
- Billing, invoicing, and payment gateway integration.
- Multi-language localization.
- Offline (disconnected) mode; the system requires connectivity at scan time.

6. Functional Requirements

6.1 Main Portal (Landing / Secure Access Gateway)

ID	Requirement	Priority
FR-01	Landing page presents the KidShield brand logo with the tagline 'Secure Access Gateway for Student Safety'.	Must
FR-02	Footer credits 'Powered by Eastern Automations'.	Must
FR-03	Parent Portal card links to registration and advertises attendance, notices, and ranks.	Must
FR-04	Staff Hub card links to the teacher console for attendance and class management.	Must
FR-05	School Admin card links to the admin console for staff, students, and system management.	Must

ID	Requirement	Priority
FR-06	Application is installable as a Progressive Web App (manifest.json + app icons).	Must

6.2 Parent Registration Portal (/register)

ID	Requirement	Priority
FR-07	Parent enters registered mobile number and requests an OTP verification code.	Must
FR-08	System simulates OTP dispatch via SMS and advances to OTP entry step.	Must
FR-09	Parent enters the 6-digit OTP; system verifies and unlocks the child registration form.	Must
FR-10	Parent provides child's full name, school name, and blood group (mandatory safety fields).	Must
FR-11	WhatsApp alert destination is pre-filled from the verified phone number and locked.	Must
FR-12	On submit, system inserts the student record and provisions an active smart tag linked to the student.	Must
FR-13	System displays a success acknowledgment screen confirming profile link and tag activation.	Must
FR-14	Parent portal provides access to child attendance, school notices, and class ranks.	Must

6.3 Staff / Teacher Hub (/teacher)

ID	Requirement	Priority
FR-15	Teacher selects an operational mode: Take Attendance, Parents Meet Option, or Lost & Found System.	Must
FR-16	System renders a live QR scanner (html5-qrcode) inside the selected mode.	Must
FR-17	Each new tag scan is appended to the current batch; duplicate scans within a batch trigger an alert.	Must
FR-18	Teacher executes batch processing; system writes attendance_logs rows per scanned tag.	Must
FR-19	PTM mode: system looks up the tag, attempts KidShield-app push notification, then falls back to WhatsApp; result logged in meeting_requests.	Must
FR-20	Lost & Found mode: system resolves the tag owner and logs a lost_found_logs entry to alert the parent.	Must
FR-21	Teacher can export the day's attendance logs as a downloadable CSV file.	Must
FR-22	Batch processing reports per-item progress and a final completion status.	Should

6.4 Principal Terminal (/principal/dashboard)

ID	Requirement	Priority
FR-23	Principal sees a live campus overview: today's attendance %, present/absent counts, total enrolled.	Must

ID	Requirement	Priority
FR-24	Classrooms tab lists class sections with attendance and assigned class-teacher contact details.	Must
FR-25	Failed Alerts tab surfaces delivery failures (e.g., invalid number) with a 'Ping Teacher' action.	Must
FR-26	Analytics tab ranks low-attendance students (action required) and the campus honor roll.	Must
FR-27	System Setup tab provides a batch-archive action for outgoing batches (e.g., graduating classes).	Must
FR-28	License module displays the campus enterprise-license expiry and a renewal action.	Should

6.5 Admin Command Center (/admin)

ID	Requirement	Priority
FR-29	Overview tab shows total enrolled, present, absent, and WhatsApp delivery-rate metrics.	Must
FR-30	Weekly attendance trend is visualized (bar chart simulation).	Must
FR-31	Live entry feed lists recent tag scans with student name, class, and time.	Must
FR-32	Staff Assignment tab manages class-teacher mapping and shows assignment status.	Must
FR-33	Unassigned classes expose an 'Assign Staff' action.	Must
FR-34	Emergency Hub provides local first-responder numbers and KidShield 24/7 support actions.	Must
FR-35	'Export Master CSV' action is available from the header.	Should

6.6 Overseer HQ Console (/hq)

ID	Requirement	Priority
FR-36	Console is gated behind an Overseer login (HQ phone number + secure PIN).	Must
FR-37	Invalid credentials produce an 'ACCESS DENIED' message and reset the passcode field.	Must
FR-38	'The QR Forge' generates a batch of tags (default 500) with quantity and target-campus inputs.	Must
FR-39	'Fleet Manager' displays global active-tag and hardware deployment analytics.	Must
FR-40	'The Vault' exposes master database CRUD access (Supabase connection placeholder in MVP).	Must
FR-41	Session can be explicitly locked out by the Overseer.	Must

6.7 Bystander Emergency Scan (/scan/[id])

ID	Requirement	Priority
FR-42	Scanning a tag URL resolves the tag UUID and loads the linked student profile.	Must
FR-43	System validates tag validity/activity and shows a clear failure state for invalid tags.	Must
FR-44	Verified view displays the child's name, school, and blood group without exposing the parent's number.	Must
FR-45	Bystander can call the parent directly via a 'Call Parent Now' action.	Must
FR-46	Bystander can message the parent via a pre-filled WhatsApp chat action.	Must

6.8 Privacy Policy (/privacy)

ID	Requirement	Priority
FR-47	Public privacy policy covers information collected, usage, sharing, security, retention, and user rights.	Must
FR-48	Policy documents compliance with the IT Act, 2000; IT (SPDI) Rules, 2011; and DPDP Act, 2023.	Must
FR-49	Policy discloses Grievance Officer contact and 48-hour acknowledgment / 30-day redressal commitment.	Must

7. Non-Functional Requirements

Category	Requirement
Performance	Dev server ready in ~1.1 s (Turbopack). Page loads under 2 s on broadband; scanner initializes within 1 s of mode selection.
Scalability	System shall support up to 2,000 enrolled students per campus with room to scale across multiple campuses (global fleet).
Security	HTTPS/TLS transmission; role-based access control; Overseer console requires clearance-level authentication; least-privilege database access.
Reliability	WhatsApp fallback ensures notification delivery when app push is unavailable; batch processing is resilient to per-item errors.
Compatibility	Responsive UI on desktop, tablet, and mobile; QR scanning requires a camera-equipped device with a modern browser.
Usability	Role portals are intuitive with clear labels, status feedback, and progress indicators for long-running operations.
Accessibility	High-contrast color schemes, semantic markup, and keyboard-navigable controls across portals.
Maintainability	Componentized React/Next.js codebase with a centralized Supabase client and mock-data boundaries for MVP.
Compliance	Alignment with the Indian IT Act, IT (SPDI) Rules 2011, DPDP Act 2023, and child-privacy principles (COPPA).

8. Business Process Flows

8.1 Parent Onboarding Flow

- 1 Parent navigates to /register and enters their registered mobile number.
- 2 System dispatches a verification OTP (simulated via SMS).
- 3 Parent enters the 6-digit OTP; the system verifies and unlocks the child profile form.
- 4 Parent submits the child's full name, school, and blood group; WhatsApp number is auto-locked.
- 5 System creates the student record and provisions an active tag in the database.
- 6 Parent receives an activation success screen; the physical tag is now live for scanning.

8.2 Daily Attendance Flow

- 1 Teacher opens the Staff Hub and selects 'Take Attendance'.
- 2 Teacher batch-scans each student's bag tag with the in-app QR scanner.
- 3 Each unique tag appends to the batch; duplicate scans are flagged.
- 4 Teacher presses 'Process N Scans'; each tag is written to attendance_logs.
- 5 CSV export is available for the day's records; Admin/Principal dashboards reflect live metrics.

8.3 Parent-Teacher Meeting Invite Flow

- 1 Teacher selects 'Parents Meet Option' and batch-scans target student tags.
- 2 System resolves each tag to the linked student and parent contact.
- 3 Attempt 1 — push notification to the KidShield parent app (if installed).
- 4 Attempt 2 — fallback WhatsApp message to the parent's verified number.
- 5 Delivery outcome is recorded in meeting_requests for tracking.

8.4 Emergency / Lost-Item Flow

- 1 A bystander scans the QR sticker on a lost bag or a child in distress.
- 2 System validates the tag and displays the verified child profile (name, school, blood group).
- 3 The bystander taps 'Call Parent Now' or 'Message via WhatsApp'.
- 4 The parent is contacted using the verified contact number; the parent's number is never displayed publicly.

8.5 Tag Provisioning Flow (Overseer)

- 1 Overseer logs in to /hq with Level-5 clearance credentials.
- 2 Overseer opens 'The QR Forge' and specifies quantity and target campus.
- 3 System generates the batch of cryptographic tag URLs and logs them securely.
- 4 Physical stickers are manufactured from the batch and distributed to campuses.
- 5 Parents activate individual tags during registration; retired batches can be archived by the Principal.

9. Data Requirements & Data Privacy

The system relies on a Supabase (PostgreSQL) backend with the following primary data entities: **students**, **schools**, **tags**, **attendance_logs**, **meeting_requests**, and **lost_found_logs**.

Entity	Key Attributes
students	id, name, school_id, class_section, blood_group, parent_whatsapp, has_parent_app, app_device_token
schools	id, name
tags	uuid, student_id, status
attendance_logs	tag_uuid, created_at
meeting_requests	tag_uuid, status
lost_found_logs	tag_uuid, created_at

Data Privacy & Protection

Child blood-group data is treated as Sensitive Personal Data or Information (SPDI) under the IT (SPDI) Rules, 2011, and is processed only with explicit parental consent. Parent phone numbers are never publicly displayed on scan pages — bystanders contact parents only through the secure in-app call/WhatsApp actions. Camera footage from QR scanning is processed locally in the browser and is never transmitted or stored by KidShield.

10. Technology Stack

Layer	Technology
Frontend	Next.js 16 (App Router), React 19, Tailwind CSS 4
PWA / Installability	Web App Manifest (manifest.json), theme-color, app icons (icon-192/512), apple-touch-icon
QR Scanning	html5-qrcode
Backend / Database	Supabase (PostgreSQL + Auth) via @supabase/supabase-js
Language	JavaScript (ES2020+), TypeScript configuration present
Linting	ESLint 9 with eslint-config-next
Build / Bundler	Turbopack (Next.js 16) for development; next build for production
Deployment Target	Vercel — production live at https://kidshield-v2.vercel.app
Environment	Node.js ≥ 20, .env.local for NEXT_PUBLIC_SUPABASE_URL / ANON_KEY

11. Assumptions, Dependencies & Constraints

11.1 Assumptions

- Every enrolled student is issued exactly one physical QR smart tag.
- Parents own a smartphone with a modern browser (and WhatsApp for fallback alerts).
- Schools have reliable internet connectivity at scan points.
- The MVP may use simulated external integrations (SMS OTP, WhatsApp gateway, push) that are stubbed in the front end.

11.2 Dependencies

- A configured Supabase project with the required tables, RLS policies, and anon key.
- Third-party notification gateways (WhatsApp Business API / Meta, Twilio) for production alerting.
- Camera permissions granted by end users in the browser for QR scanning.

11.3 Constraints

- v2.0 is web-only; no native mobile apps or offline mode.
- The Overseer database vault currently depends on a future Supabase connection.
- Simulated integrations must be replaced with real gateways before broad production rollout.

12. Risks & Mitigation Strategies

Risk	Severity	Mitigation
Duplicate or fraudulent tag scans	Medium	Duplicate detection in scanner; per-batch state; tag status lifecycle (active/archived).
Notification delivery failure	Medium	Waterfall routing: app push → WhatsApp fallback → logged failure surfaced to Principal.
SPDI data exposure (blood group)	High	Explicit consent; never display parent numbers; encrypted transport; role-based access.
Camera permission / hardware issues	Medium	Graceful scanner cleanup; clear UI feedback; manual fallback entry planned.
Supabase connectivity outages	Medium	Per-item error resilience in batch processing; clear error messages.
Unauthorized HQ access	High	Level-5 clearance login gate; session lock-out; audit logging of IP.

13. Release Phases & Implementation Roadmap

Phase	Scope	Status
Phase 1 — MVP (current)	Deployed live at kidshield-v2.vercel.app; PWA installability, Eastern Automations branding, redesigned portals, phone+PIN HQ auth, real-time dashboard sync, system-status panel.	Live
Phase 2 — Pilot	Single-campus pilot; real WhatsApp gateway; principal & admin dashboards connected to live data.	Next
Phase 3 — Production	Full notification stack (SMS, push, WhatsApp); Overseer vault connected; audit & compliance hardening.	Planned
Phase 4 — Scale	Multi-campus fleet analytics; native mobile apps; offline scanning mode; localization.	Future

14. Key Performance Indicators (KPIs)

KPI	Target
Average attendance-processing time per class	< 60 seconds
Attendance accuracy	≥ 99%
PTM notification delivery rate	≥ 98%
Parent onboarding completion	≥ 90% of enrolled students
Lost-item resolution time	≤ 24 hours
System uptime	≥ 99.5% during school hours
Scan-page load time	< 2 seconds

15. Approval & Sign-off

By signing below, the undersigned stakeholders acknowledge that they have reviewed this Business Requirement Document and agree that it accurately captures the business needs for the KidShield v2.0 ecosystem.

Role / Title	Name	Signature	Date
Product Owner			
School Administrator			
Principal Representative			
Engineering Lead			
QA / Acceptance Lead			

Document Version History

Version	Date	Change Description	Author
1.0	02 Aug 2026	Initial business requirement baseline for stakeholder review.	Product Management
1.1	02 Aug 2026	Aligned with deployed live site (kidshield-v2.vercel.app): PWA support, Eastern Automations branding, redesigned portals, phone+PIN HQ auth, real-time sync, system-status panel.	Product Management